

CS5187 Final Exam Review / 期末复习资料

Course / 课程: CS5187: Vision and Image Instructor / 授课教师: Prof. Zhu Zhiyu

1. Knowledge Framework / 知识框架

Part I: Low-Level Vision / 低层视觉 (lec00–lec05)

Lecture	Topic / 主题
lec00	Introduction to Computer Vision / 计算机视觉导论
lec01	Images & Image Filtering / 图像与图像滤波
lec02	Edge Detection / 边缘检测
lec03	Image Resampling & Interpolation / 图像重采样与插值
lec04	Harris Corner Detection / Harris角点检测
lec05	Feature Invariance / 特征不变性

Part II: Geometry & Appearance / 几何与外观 (lec06–lec10, lec13–lec14)

Lecture	Topic / 主题
lec06	Feature Descriptors & Matching / 特征描述子与匹配
lec07	Image Transformations & Warping / 图像变换与变形
lec08	Image Alignment / 图像对齐
lec09	RANSAC / 随机抽样一致
lec10	Cameras & Projection / 相机与投影
lec13	Binocular Stereo / 双目立体视觉
lec14	Multi-View Stereo / 多视图立体

Part III: Recognition & Deep Learning / 识别与深度学习 (lec15–lec16)

Lecture	Topic / 主题
lec15	Neural Networks / 神经网络
lec16	CNNs, RNNs, Attention & Transformers / 卷积网络、循环网络、注意力与Transformer

Part IV: Generative Models / 生成模型 (下 semester)

Lecture	Topic / 主题
lecure4	Maximum Likelihood & MLE / 最大似然估计
lecure5	Latent Variable Models & ELBO / 潜变量模型与证据下界
LVM-02	VAE & Reparameterization / 变分自编码器与重参数化
NF-01	Normalizing Flow Models (1) / 标准化流模型(1)
lecure8	Normalizing Flow Models (2) / 标准化流模型(2)
gan9s	GANs / 生成对抗网络
lecure10	I-GANs, Wasserstein GAN, BIGAN, CycleGAN / f-散度GAN, WGAN, 双向GAN, 循环GAN
lecure11	Energy-Based Models / 基于能量的模型
lecure12	Score Matching & NCE / 分数匹配与噪声对比估计
lecure13	Score-Based Models (1) / 基于分数的模型(1)
lecure14	Score-Based Models (2) / 基于分数的模型(2)
lecure15	Diffusion Models (DDPM, SDE) / 扩散模型
lecure16	Diffusion Models: SDE+ODE, Architecture / 扩散模型进阶

2. Core Concepts / 核心概念梳理

2.1 Image Filtering / 图像滤波

Convolution / 卷积

- Definition (定义): Replace each pixel with a weighted sum of its neighbors using a kernel. / 用卷积核将每个像素替换为其邻域的加权求和。
- Formula (公式): $G[i,j]=\sum H[m,n] \cdot F[i-m,j-n]$
- Key property (关键性质): Convolution is associative and commutative; convolution with self is another Gaussian. / 卷积满足结合律和交换律。

Gaussian Filter / 高斯滤波器

- Definition (定义): A low-pass filter that removes high-frequency components. / 去除高频成分的低通滤波器。
- Property (性质): Convolving two Gaussians of σ_1 and σ_2 yields a Gaussian of $\sigma=\sqrt{\sigma_1^2+\sigma_2^2}$. / 两个高斯卷积的结果仍为高斯。

Sharpening Filter / 锐化滤波器

- Definition (定义): $\text{original} + (\text{original} - \text{blurred}) = 2 \cdot \text{original} - \text{blurred}$. / 原图加上(原图减模糊图)实现锐化。
- Also called (亦称): High-pass filter / 高通滤波器。

2.2 Edge Detection / 边缘检测

Image Gradient / 图像梯度

- Definition (定义): $\nabla f=[\partial f/\partial x,\partial f/\partial y]$, points in direction of most rapid increase. / 指向强度增长最快的方向。
- Gradient magnitude (梯度幅值): $\|\nabla f\| = (\partial f/\partial x)^2+(\partial f/\partial y)^2$
- Gradient direction (梯度方向): $\theta=\text{atan2}(\partial f/\partial y,\partial f/\partial x)$

Sobel Operator / Sobel算子

- Definition (定义): Common approximation of derivative of Gaussian. / 高斯导数的常用近似。
- Kernels (核):
 - x-direction: $\begin{bmatrix} -1, 0, 1 \\ -2, 0, 2 \\ -1, 0, 1 \end{bmatrix}$
 - y-direction: $\begin{bmatrix} 1, 2, 1 \\ 0, 0, 0 \\ -1, -2, -1 \end{bmatrix}$

Canny Edge Detector / Canny边缘检测器

1. Filter image with derivative of Gaussian / 用高斯导数滤波
2. Find magnitude and orientation of gradient / 求梯度幅值和方向
3. Non-maximum suppression / 非极大值抑制
4. Hysteresis thresholding (two thresholds: high T, low t) / 滞后阈值化
5. Strong edges: $R > T$; Weak edges: $t < R < T$ (kept only if connected to strong edge) / 强边缘保留，弱边缘仅在它与强边缘相连时保留

Scale Space / 尺度空间

- Properties (性质): Edge position may shift with σ ; two edges may merge; an edge cannot split into two. / 边缘位置可能随 σ 偏移，可合并但不能分裂。

2.3 Image Resampling / 图像重采样

Aliasing / 混叠

- Definition (定义): Occurs when sampling rate < $2 \times$ max frequency (Nyquist rate). / 采样率低于奈奎斯特率时发生。
- Solution (解决方案): Gaussian pre-filtering before subsampling. / 下采样前先做高斯预滤波。

Gaussian Pyramid / 高斯金字塔

- Definition (定义): Repeatedly blur and downsample by $2 \times$. 反卷积模糊后2倍下采样。
- Space (空间): Total space = $4/3 \times$ original image. / 总空间为原图的 $4/3$ 。

Interpolation Methods / 插值方法

- Nearest-neighbor (最近邻): Fastest, but blocky. / 最快但产生方块效应。
- Bilinear (双线性): Better quality, tent function. / 质量更好。
- Bicubic (双三次): Best quality among common choices. / 常用方法中质量最好。

2.4 Harris Corner Detection / Harris角点检测

SSD Error / SSD误差

- Formula (公式): $E(u,v)=\sum Ww(x,y)[(x+u,y+v)-I(x,y)]^2$
- Small motion approximation (小运动近似): $E(u,v)=[u \ v] \cdot H \begin{bmatrix} u \\ v \end{bmatrix}$, where H is the second moment matrix. / 二阶矩矩阵H近似。

Second Moment Matrix / 二阶矩矩阵

- Definition (定义): $H=\sum w(x,y)[lx^2lxyly^2]$
- Eigenvalues (特征值): λ_1, λ_2 determine corner/edge/flat classification. / 决定角点/边缘/平区域分类。

Classification / 分类

Case (情况)	λ_1, λ_2	Classification (分类)
Both small (都很小)	$\lambda_1=\lambda_2=0$	Flat / 平坦区域
One large, one small (一大一小)	$\lambda_1 \gg \lambda_2$ or $\lambda_2 \gg \lambda_1$	Edge / 边缘
Both large (都很大)	$\lambda_1-\lambda_2 \gg 0$	Corner / 角点

Harris Operator / Harris算子

- Formula (公式): $R=\det(H)-\alpha \text{trace}(H)^2+\lambda_1 \lambda_2-\alpha(\lambda_1+\lambda_2)^2$
- Typical α (典型参数): $\alpha \in [0.04, 0.06]$

Harris Detector Steps / Harris检测步骤

1. Compute image gradients / 计算图像梯度
2. Square of derivatives, Gaussian filter / 梯度平方后高斯滤波
3. Compute cornerness function / 计算角点响应函数
4. Non-maxima suppression / 非极大值抑制

2.5 Feature Invariance / 特征不变性

Invariance vs. Equivariance / 不变性与等变性

- Invariance (不变性): Feature locations don't change after transformation. / 变换后特征位置不变。
- Equivariance (等变性): Features detected at corresponding locations in transformed images. / 变换后特征在对应位置被检测到。

Harris Invariance for Properties / Harris不变性

Transformation (变换)	Property (性质)
Translation (平移)	Equivalent / 等变
Rotation (旋转)	Equivalent (eigenvalues unchanged) / 等变
Affine intensity (仿射强度)	Partially invariant / 部分不变
Scale (尺缩)	NOT invariant / 不变

Scale Invariant Detection / 尺度不变检测

- Key idea (核心思想): Find local maxima in both position AND scale. / 在位置 and 尺度上同时寻找局部极大值。
- Laplacian of Gaussian (LoG): Blob detector, find maxima in position-scale space. / 斑点检测器，在位置-尺度空间找极大值。
- Difference of Gaussians (DoG): Approximation of LoG, used in SIFT. / LoG的近似，SIFT中使用。

Characteristic Scale / 特征尺度

- Definition (定义): Scale producing peak LoG response. / 产生LoG峰响应的尺度。

2.6 Feature Descriptors & Matching / 特征描述子与匹配

SIFT Descriptor / SIFT描述子

- Process (过程):
 - Take 16×16 window around feature / 取 16×16 窗口
 - Compute gradient orientation per pixel / 计算每像素梯度方向
 - Threshold weak edges / 阈值化弱边缘
 - Create 4×4 grid of cells, 8 orientation bins each / 4×4 网格，每格8方向bin

- Result: 128-dimensional descriptor / 结果为128维描述子
- Properties (性质): Robust to viewpoint (up to $\sim 60^\circ$), illumination changes. / 对视点变化($\sim 60^\circ$ 以内) 和光照变化鲁棒。

HOG Descriptor / HOG描述子

- Steps: Gradient computation $\rightarrow$ Cell histograms (9 bins) $\rightarrow$ Block normalization (L2-norm). / 梯度计算 $\rightarrow$ 单元直方图 $\rightarrow$ 块归一化。

ORB Descriptor / ORB描述子

- Steps: oFAST keypoint detection $\rightarrow$ rBRIEF binary descriptor (256 bits). / oFAST关键点检测 $\rightarrow$ rBRIEF二值特征描述子。

Feature Matching / 特征匹配

- L2 distance: $\|f_1-f_2\|$ / $\|f_{\{1\}}-f_{\{2\}}\|$ ($f_{\{2\}}$ = best match, $f_{\{2\}}'$ = 2nd best) / 比值为距离
- Ratio distance: $\|f_1-f_2\| / \|f_{\{1\}}-f_{\{2\}}\|$ ($f_{\{2\}}$ = best match, $f_{\{2\}}'$ = 2nd best) / 比值距离
- ROC Curve: True positive rate vs. false positive rate; AUC summarizes performance. / ROC曲线与AUC指标。

2.7 Image Transformations / 图像变换

Transformation Hierarchy / 变换层级

Transform (变换)	Matrix (矩阵)	DOF	Properties (性质)
Translation (平移)	$\begin{bmatrix} x & y \\ 0 & 1 \end{bmatrix}$	2	Origin may shift / 原点可移动
Euclidean (欧氏)	$R + t$	3	Preserves distances / 保持距离
Similarity (相似)	$sR + t$	4	Preserves angles / 保持角度
Affine (仿射)	$\begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$	6	Parallel lines preserved / 平行线保持
Homography (射影/射影几何)	General 3×3	8	Lines preserved, parallel not / 直线保持，平行不一定

Homography / 单应变换

- Definition (定义): 8-DOF projective transformation; defined up to scale. / 8自由度射影变换，由尺度因子决定。
- Minimum matches (最少匹配): 4 point correspondences. / 4对匹配点。

Forward vs. Inverse Warping / 前向与逆向变形

- Forward warping: Send source pixels to target; may cause holes. / 源像素送到目标，可能产生空洞。
- Inverse warping: For each target pixel, sample from source via $T^{-1}(\cdot)$; use interpolation. / 对每个目标像素从源采样，需插值。

2.8 Image Alignment & Least Squares / 图像对齐与最小二乘

Least Squares / 最小二乘法

- Translation (平移): $t^* = \text{mean displacement}$; normal equations: $ATA=tATb$. / 平移的最优解为平均位移。
- Affine (仿射): 6 unknowns, need ≥ 3 matches. / 6个未知量，至少3对匹配。
- Homography (单应): Solve via SVD; $h = \text{eigenvector of } ATA \text{ with smallest eigenvalue}$. / 通过SVD求解。

2.9 RANSAC / 随机抽样一致

Algorithm / 算法步骤

1. Randomly select s samples (minimum to fit model) / 随机选 s 个样本
2. Fit model / 拟合模型
3. Count inliers (within threshold ϵ) / 计数内点
4. Repeat N times / 重复N次
5. Choose model with most inliers / 选择内点最多的模型
6. Refit using all inliers (least squares) / 用所有内点重新拟合

Number of Rounds / 迭代次数

- Formula (公式): $N=\log(1-p)/\log(1-(1-e)^s)$
- $p = \text{desired probability of success / 期望成功概率}$
- $e = \text{outlier ratio / 外点比例}$
- $s = \text{minimum sample size / 最小样本数}$

Pros & Cons / 优缺点

- Pros: Simple, general, often works well. / 简单、通用、效果好。
- Cons: Parameters to tune; can fail for very low inlier ratios. / 需调参；内点率极低时可能失败。

2.10 Cameras & Projection / 相机与投影

Pinhole Camera / 针孔相机

- Projection (投影): $(x,y,z) \rightarrow (f+xz/f, yz/f)$ using similar triangles. / 利用相似三角形投影。

- Perspective projection matrix / 透视投影矩阵: $K = \begin{bmatrix} f & 0 & cx \\ 0 & f & cy \\ 0 & 0 & 1 \end{bmatrix}$

Camera Parameters / 相机参数

- Intrinsic (内参): focal length f , principal point (cx,cy) , aspect ratio α , skew s . / 焦距、主点、长宽比、倾斜。
- Extrinsic (外参): Rotation R (3×3) and translation t (3×1). / 旋转矩阵和平移向量。
- Full projection (完整投影): $x_{\text{pixel}}=K[R \ t]X$

Projection Types / 投影类型

- Perspective (透视): General case, parallel lines converge. / 一般情况下，平行线汇聚。
- Orthographic (正交): Distance to image plane ∞ ; $(x,y,z) \rightarrow (x,y)$. / 距离无限远。
- Weak perspective (弱透视): Scaled orthographic; good for small depth range. / 缩放正交投影。

Radial Distortion / 径向畸变

- Types: Barrel (桶形) vs. Pincushion (枕形)
- Model: $r_{\text{corrected}}=r(1+k_1r^2+k_2r^4+\dots)$

2.11 Stereo Vision / 立体视觉

Disparity & Depth / 视差与深度

- Disparity (视差): $d=x_1-x_2$ (horizontal pixel difference between views). / 左右视图水平像素差。
- Depth (深度): $Z=fB/d$, where $B = \text{baseline}$, $f = \text{focal length}$. / 深度与视差成反比。

Stereo Matching / 立体匹配

- SSD matching: Sum of squared differences within a window. / 窗口内平方差之和。
- Window size tradeoff: Small $\rightarrow$ more detail but more noise; Large $\rightarrow$ less noise but less detail. / 窗口大小的权衡。
- Stereo as Energy Minimization / 立体视觉作为能量最小化
 - Energy: $E(d)=\sum data_cost(x,d(x))+\sum smoothness_cost(d(x),d(y))$
 - Smoothness costs: Potts model (0 if same, λ if different); L1 distance. / Potts模型：1距离。

Multi-View Stereo (MVS) / 多视图立体

- SSDD: Sum of SSD over multiple baselines. / 多基线SSD之和。
- Matching scores: SSD, SAD, ZNCC (zero-mean normalized cross-correlation). / 匹配评分方法。

Plane-Sweep Stereo / 平面扫描立体

- Sweep family of planes at different depths, reproject neighbors, compare. / 在不同深度扫描平面，重投影邻居并比较。
- NERF: Represent scenes as $(x,y,z) \rightarrow \text{RGB}+\alpha$ functions, volume rendering. / 用神经网络表示场景的辐射场。

2.12 Neural Networks / 神经网络

Linear Classifier / 线性分类器

- Score function (得分函数): $f(x;W)=Wx+b$
- Three viewpoints (三个视角): Algebraic (代数), Geometric (几何), Template matching (模板匹配)
- Loss Functions / 损失函数
 - Cross-entropy loss (交叉熵损失): $L=-\log(\exp(f(y))/\sum \exp(f(j)))$
 - Softmax (软最大): $P(y=j|x)=\exp(f(j))/\sum \exp(f(k))$

Neural Network Architecture / 神经网络架构

- MLP (多层感知机): Linear functions separated by non-linear activations. / 非线性函数插入非线性激活函数。
- Gradient descent (梯度下降): $W \leftarrow W-\eta \nabla L(W)$; learning rate η crucial. / 学习率 η 至关重要。
- Backpropagation (反向传播): Efficient gradient computation via chain rule. / 通过链式法则高效计算梯度。

2.13 CNNs, RNNs, Attention, Transformers / CNN、RNN、注意力、Transformer

Convolutional Layer / 卷积层

- Output size (输出尺寸): $(N-F)/\text{stride}+1$
- Filters (滤波器): Extend full depth of input; produce activation maps. / 滤波器覆盖输入全深度，产生激活图。
- Padding & Stride: Control output spatial dimensions. / 控制输出空间维度。

Pooling Layer / 池化层

- Max pooling (最大池化): Reduces spatial dimensions; operates per activation map. / 减小空间维度，逐激活图操作。

RNN / 循环神经网络

- Recurrence (递推): $h_t=W(x_t+U \cdot h_{t-1})$; same weights at each time step. / 每个时间步使用相同权重。
- BPTT: Backpropagation Through Time. / 时间反向传播。
- Truncated BPTT: Backpropagate only for limited steps. / 截断时间反向传播。
- Types: Vanilla RNN, LSTM, GRU. / 类型。

Self-Attention / 自注意力

- Query-Key-Value (查询-键-值): Attention(Q,K,V)=softmax(QKT/dkV)
- Multi-Head Attention (多头注意力): Multiple attention heads in parallel; concatenate outputs. / 多个注意力头并行，拼接输出。

Transformer

- Encoder (编码器): Self-attention + FFN, with residual connections and layer norm. / 自注意力+前馈网络，带残差连接和层归一化。
- Decoder (解码器): Masked multi-head attention (cannot see future) + cross-attention + FFN. / 掩码多头注意力+交叉注意力+前馈网络。
- Positional Encoding (位置编码): Injects sequence order information. / 注入序列顺序信息。

2.14 Maximum Likelihood Estimation / 最大似然估计

KL Divergence / KL散度

- Definition (定义): $D_{KL}(p||q)=\mathbb{E}_p[-\log(p(x)/q(x))]$ ≥ 0 ; asymmetric. / 非对称，衡量两个分布的差异。
- Interpretation (解释): Extra bits needed when using code optimized for q on data from p . / 用 q 的编码码字来自 p 的数据时多用的比特数。

MLE / 最大似然估计

- Formula (公式): $\theta^*=\text{argmax}_{\theta} \mathbb{E}[\log P_\theta(x)]$

- Equivalence (等价性): Minimizing $KL(p_{data}||p_\theta) \equiv \text{maximizing expected log-likelihood}$. / 最小化KL散度等价于最大化期望对数似然。

Monte Carlo Estimation / 蒙特卡洛估计

- Unbiased (无偏): $\mathbb{E}[g^s]=\mathbb{E}[g(x)]$
- Variance (方差): $\text{Var}(g^s)=\text{Var}(g(x))/T$; reduces with more samples. / 更多样本降低方差。

Bias-Variance Tradeoff / 偏差-方差权衡

- High bias (高偏差): Underfitting (model too simple). / 欠拟合。
- High variance (高方差): Overfitting (model too complex). / 过拟合。
- Regularization (正则化): objective=loss+ $\lambda R(M)$. / 目标函数加入正则项。

2.15 Latent Variable Models & VAE / 潜变量模型与变分自编码器

Latent Variable Models / 潜变量模型

- Generative process (生成过程): $z \sim p(z)$, $x \sim p(x|z)$. / 先采样潜变量，再由其生成数据。
- Marginal likelihood (边际似然): $p(x)=\int p(x,z)p(z)dz$ — often intractable. / 通常难以计算。

Mixture of Gaussians / 高斯混合模型

- $z \sim \text{Categorical}(\pi_1, \dots, \pi_K)$; $p(x|z,k)=\mathcal{N}(x|\mu_k, \Sigma_k)$. / 离散潜变量的混合模型。

VAE / 变分自编码器

- Infinite mixture (无限混合): $z \sim \mathcal{N}(0,1)$; $p(x|z)=\mathcal{N}(\mu(z), \Sigma(z))$; μ, Σ are neural networks. / 连续潜变量的无限混合高斯。

ELBO / 证据下界

- Formula (公式): $\log p(x) \geq \mathbb{E}_z[\log p(x|z)] + H(q||p)$ ($q=p(z)$)
- Equality condition (等号条件): When $q||p$: $\text{warp}(p) = p(z|x|\text{theta})$, i.e., $KL(q||p(z|x))=0$. / 当分布分布等于真实后验时等号成立。
- Decomposition (分解): $\log p(x)=ELBO+DKL(q(z)||p(z|x))$

Autoencoder Perspective / 自编码器视角

- ELBO = Reconstruction term + KL term: $L=L_{\text{eq}}(q(z)||p(z))-DKL(q(z)||p(z))$
- Reconstruction (重建项): Encourages $x^*=x$. / 鼓励重建接近原始输入。
- KL term (KL项): Forces latent distribution close to prior $p(z)$. / 迫使潜变量分布接近先验。

Reparameterization Trick / 重参数化技巧

- Key idea (核心思想): Sample $z \sim \mathcal{N}(0,1)$, then $z=\mu+\sigma\epsilon$. / 将随机性转移到 ϵ ，使梯度可通过确定性函数传递。
- Benefit (优势): Enables gradient-based optimization; lower variance than REINFORCE. / 实现梯度优化，方差低于REINFORCE。

Amortized Inference / 摊销推断

- Idea (思想): Learn $q_\phi(z|x)=\text{encoder}(x)$ to map each x to variational parameters. / 学习编码器网络将输入映射到变分参数，避免逐点优化。

2.16 Normalizing Flow Models / 标准化流模型

Core Idea / 核心思想

- Make $y=f_\theta(x)$ deterministic and invertible; then $p(y)$ is tractable via change of variables. / 使映射确定且可逆，通过变量替换公式计算似然。

Change of Variables / 变量替换公式

- Advantages over original GAN (相比原始GAN的优势):
- Loss correlates with sample quality / 损失与样本质量相关
- More stable training / 训练更稳定
- No mode collapse in practice / 实践中无模式坍塌

2.20 BiGAN / 双向生成对抗网络

Bidirectional GAN (BiGAN) / 双向GAN

- Motivation (动机): Standard GANs learn a generator $G:z\rightarrow x$ but cannot infer the latent representation z given x / 标准GAN只学到生成器，无法推断潜变量表示。
- Architecture (架构): Add an encoder $E:x\rightarrow z$ alongside the generator $G:z\rightarrow x$; discriminator D takes (x,z) pairs and outputs $s(x,E(x))$ from $(G(z),z)$ / 增加编码器E，判断器在 (x,z) 对上区分编码与生成对。
- Objective (目标): $\min_G, \max_D V(D,E,G)=E_x-E_{pdata}[\log D(E(x))]+E_{z\sim p(z)}[\log(1-D(G(z),x))]$ /
- Result (结果): At equilibrium, $E=E_G^{-1}$ (encoder inverts the generator) / 均衡时编码器为生成器的逆。
- ALI (Adversarially Learned Inference): Simultaneous proposal, equivalent to BiGAN / 同时提出的等价方法。

2.21 CycleGAN / 循环生成对抗网络

Image-to-Image Translation / 图像到图像翻译

- Problem (问题): Given images from two domains X and Y , learn to translate between them. / 给定两个域的图像，学习在域间翻译。
- Paired vs. Unpaired (配对与非配对): Paired examples are expensive; CycleGAN works with unpaired data. / 配对数据昂贵，CycleGAN使用非配对数据。

CycleGAN Architecture / CycleGAN架构

- Generators (生成器): $G:X\rightarrow Y$ and $F:Y\rightarrow X$ ，/ 双向生成器。
- Discriminators (判别器): D_Y distinguishes real Y from $G(X)$; D_X distinguishes real X from $F(Y)$ / 双向判别器。
- Adversarial loss (对抗损失): $z_{GAN}(G,D_Y,X)+z_{GAN}(F,D_X,Y)$ / 双向对抗损失。
- Cycle consistency loss (循环一致性损失): $\lambda[E(X)\parallel F(G(X))-X]+1+E(Y)\parallel G(F(Y))-Y\parallel]$ / 确保往返翻译一致。
- Overall objective (总目标): $\min(F,G,D_X,D_Y)\lambda z_{GAN}+\lambda z_{cyc}$.

AlignFlow / 对齐流

- Key idea (核心思想): If G is a normalizing flow model, $F=G^{-1}$ is free; exactly cycle-consistent. / 若生成器为流模型，逆向自然可得，精确循环一致。
- Training (训练): Can train via MLE and/or adversarial learning. / 可通过MLE和/或对抗学习训练。

StarGAN / 星形GAN

- Extension (扩展): Handles multiple domains with a single generator. / 单一生成器处理多个域的翻译。

2.22 Energy-Based Models / 基于能量的模型

EBM Formulation / EBM公式

- Definition (定义): $p(\theta)=\exp(f(\theta))/Z(\theta)=1/Z(\theta)\exp(f(\theta))$, where $f(\theta)$ is any function (e.g., neural network) and $Z(\theta)=\int \exp(f(\theta))dx$ is the partition function. / 任意函数取指数后归一化，分母为配分函数。
- Key property (关键性质): Very flexible architecture – any $f(\theta)$ can be any neural network (no invertibility, factorization, or latent variable constraints). / 架构极其灵活， $f(\theta)$ 可为任意神经网络。
- Partition function (配分函数): $Z(\theta)=\int \exp(f(\theta))dx$ – generally intractable. / 通常不可计算。

Energy Function / 能量函数

- Interpretation (解释): $-f(\theta)$ is the "energy"; low energy = high probability. / $-f(\theta)$ 为“能量”，低能量对应高概率。
- Non-negativity trick (非负技巧): Given any function $f(\theta)$, ensure non-negativity via $g(\theta)=f(\theta)/x^2$, $\exp(f(\theta))$, $f(\theta)/x!$, or $\log(1+\exp(f(\theta)/x))$ / 通过平方、指数、绝对值等保证非负。

Maximum Likelihood Training / 最大似然训练

- Log-likelihood (对数似然): $\log p(\theta)=f(\theta)/Z(\theta)$ / 对数似然等于函数值减去配分函数对数。
- Gradient (梯度): $\nabla \log p(\theta_{train})=\nabla \theta f(\theta_{train})-E_x-p\theta[\nabla f(\theta,x)]$ / 梯度=正相位（维高训练数据能量）减负相位（拉低模型采样能量）。
- Positive phase vs. Negative phase (正相位与负相位):
- Positive phase (正相位): $\nabla \theta f(\theta_{train})$ – increases probability of training data. / 提高训练数据概率。
- Negative phase (负相位): $E_x-p\theta[\nabla f(\theta,x)]$ – decreases probability of model samples. / 降低模型样本概率。

Contrastive Divergence / 对比度数

- Algorithm (算法): Instead of computing $Z(\theta)$ exactly, use Monte Carlo estimate. / 用蒙特卡罗估计代替精确计算配分函数。
- Update step (更新步骤): $\nabla \theta f(\theta_{train})-\nabla \theta f(\theta_{sample})$, where $\theta_{sample}\sim p\theta$. / 用模型采样的梯度近似负相位。

- Key idea (核心思想): Make training data more likely than typical samples from the model. / 使训练数据比模型典型样本概率更高。

Sampling from EBMs / 从EBM采样

- MCMC (马尔可夫链蒙特卡罗):
- Initialize x_0 randomly / 随机初始化
- Propose $x_{t+1}\sim\alpha(\text{noise})$ / 提议新样本
- If $f(\theta_t)>f(\theta(x_{t+1}))$, accept; else accept with probability $\exp(f(\theta_t)-f(\theta(x_{t+1})))$ / 接受或以一定概率拒绝

- Converges to $p\theta$, but can be slow / 收敛但可能很慢
- Langevin MCMC (朗之万MCMC): $xt+1=xt+\epsilon \nabla x \log p(\theta_t)x+\epsilon z_{2\epsilon t}$, where $z_t\sim N(0,1)$. / 利用梯度信息加速采样的MCMC方法。
- Score function (分数函数): For EBMs, $\nabla \log p(\theta,x)=\nabla f(\theta,x)$ (since $\nabla \log Z(\theta)=0$). / EBM的分数函数（对数密度对 x 的梯度）恰好等于 f 的梯度。

Score-Based Generative Modeling / 基于分数的生成模型

- Key idea (核心思想): Directly estimate the score function $\nabla \log p(x)$ using a neural network $s(\theta)=\nabla \log p(x)$ / 直接用神经网络估计分数函数。
- Score matching (分数匹配): Train $s\theta$ by minimizing $E_{pdata}[\parallel s\theta(x)-\nabla \log p_{data}(x)\parallel^2]$ (or variants avoiding $\nabla \log p_{data}$). / 通过分数匹配目标训练。
- Denoising Score Matching (去噪匹配): Add noise z to data, then match the score of the perturbed distribution. / 对数据加噪声后匹配扰动分布的分数。
- Noise Conditional Score Network (NCSN): Use multiple noise levels $\{\sigma_1>\sigma_2>\cdots>\sigma_L\}$; train one network conditioned on noise level. / 使用多尺度噪声，训练条件分数网络。
- Annealed Langevin Dynamics (退火朗之万动力学): Sequentially sample with decreasing noise levels; each level initialized from previous. / 逐步降低噪声水平采样。

EBM vs. Other Generative Models / EBM与其他生成模型比较

- vs. Autoregressive/Flow/VAE: EBM has much more flexible architecture (no restrictions on $p\theta$), but sampling is slower (requires MCMC). / EBM架构更灵活，但采样需要MCMC。
- vs. GAN: EBM provides stable training and tractable (unnormalized) likelihood; GAN has fast sampling but unstable training and intractable likelihood. / EBM训练稳定，GAN采样快但训练不稳定。
- Composability (可组合性): $p_{combine}(x)\propto \Pi p_i(x|a_i)$ – can combine multiple EBMs by multiplying their densities. / 多个EBM可乘性组合，适用于零样本组合推理。

3. Key Formulas & Theorems / 重点公式与定理

3.1 Convolution / 卷积

Formula: $G[y]_i=\sum_m\sum_n f[m,n]\cdot f[i-m,j-n]$

Variables: H = kernel, F = input image, G = output image
Conditions: Kernel size determines neighborhood / 核大小决定邻域范围

3.2 Gaussian Filter / 高斯滤波

Formula: $G(x,y)=1/2\pi\sigma^2\exp(-(x+y)^2/2\sigma^2)$

Variables: σ =sigma = standard deviation (scale) / 标准差（尺度）
Conditions: Larger σ = more blurring / σ 越大越模糊

3.3 Image Gradient / 图像梯度

Formula: $\nabla f=[\partial f/\partial x,\partial f/\partial y]$; $\|\nabla f\|=(\partial f/\partial x)^2+(\partial f/\partial y)^2$

Variables: ∂ =partial/partialx, ∂ =partial/partialy = partial derivatives / 偏导数
Conditions: Computed via finite differences or derivative of Gaussian / 通过有限差分或高斯导数计算

3.4 Harris Corner Response / Harris角点响应

Formula: $R=\det(H)$ -atrace(H)-2 $\lambda_1\lambda_2$ - $\alpha(\lambda_1+\lambda_2)^2$

Variables: H = second-order moment matrix, $\alpha\in[0.04,0.06]$, λ_1,λ_2 = eigenvalues
Conditions: $R>0$ = corner; $R=0$ = edge; $R<0$ = flat (using alternate sign convention)

3.5 Perspective Projection / 透视投影

Formula: $(x_c,y_c,z_c)\rightarrow(fx/z,cx,fy/z+cy)$

Variables: f = focal length, (cx,cy) = principal point
Conditions: Pinhole camera model / 针孔相机模型

3.6 Stereo Depth / 立体深度

Formula: $Z=d\cdot B/d$

Variables: Z = depth, f = focal length, B = baseline, d = disparity
Conditions: Rectified stereo pair / 校正后的立体对

3.7 RANSAC Iterations / RANSAC迭代次数

Formula: $N=\log(1-p)/\log(1-e)$

Variables: p = success probability, e = outlier ratio, s = min sample size
Conditions: Assumes $<50\%$ outliers / 假设外点比例 $<50\%$

3.8 ELBO / 证据下界

Formula: $\log p(x;\theta)=\mathbb{E}_q[\log p(x,z;\theta)]-E_q[\log p(z;\theta)]=E_q[\log p(z|x;\theta)]-DKL(q\phi(z|x)\parallel p(z))$

Variables: q = variational distribution, θ = model parameters, ϕ = variational parameters
Conditions: Equality when $q(z|\text{varphi})=p(z|x;\theta)$ / 当变分分布等于后验时等号成立

3.9 Change of Variables (Flow) / 变量替换（标准流化）

Formula: $p_X(x)=p_Z(f^{-1}(x))\cdot|\det(\partial f^{-1}/\partial x)|$

Variables: f = invertible transformation, $x,z\in\mathbb{R}^n$ (same dimension)
Conditions: f must be invertible; ∂f continuous / f 必须可逆； x,z 连续且同维

3.10 GAN Objective / GAN目标

Formula: $\min_G\max_D-E_{pdata}[\log D(x)]+E_{z\sim p(z)}[\log(1-D(G(z)))]$

Variables: G = generator, D = discriminator, p = prior on z

Conditions: At optimum, differentiating JSD / 最优时等价于最小化JS散度

3.11 Cross-Entropy Loss / 交叉熵损失

Formula: $L_i=-\log(\exp(f(y_i))/Z(\exp(f)))$

Variables: f_j = score for class j ; y_i = true label

Conditions: Most common classification loss / 最常用的分类损失

3.12 Conv Output Size / 卷积输出尺寸

Formula: $\text{output}=(N+2F-2P)\text{stride}+1$

Variables: N = input size, F = filter size, P = padding
Conditions: Must be integer / F 必须为整数

3.13 f-Divergence / f-散度

Formula: $D^f(p,q)=E_x-q[f(p(x)q(x))]$, where f is convex with $f(1)=0$

Variables: p,q = probability distributions, f = convex generator function

Conditions: $D^f(p,q)>0$, equals 0 iff $p=q$ / 非负，等号当且仅当两分布相同

3.14 Wasserstein Distance / Wasserstein距离

Formula: $W(p,q)=\sup\{\parallel t\parallel_1\leq 1|E_x-\text{pr}(f(x))=E_y-\text{pr}(g(y))\}$

Variables: p,q = real and generated distributions, $f=1$ -Lipschitz function

Conditions: Meaningful even for non-overlapping distributions / 即使分布不重叠也有意义

3.15 EBM Probability / EBM概率

Formula: $p(\theta)=1/Z(\theta)\exp(f(\theta))$, where $Z(\theta)=\int \exp(f(\theta))dx$

Variables: $f(\theta)$ = any function (energy negative), $Z(\theta)$ = partition function

Conditions: $Z(\theta)$ generally intractable / 配分函数通常不可计算

3.16 EBM Log-Likelihood Gradient / EBM对数似然梯度

Formula: $\nabla \log p(\theta_{train})=\nabla \theta f(\theta_{train})-E_{x\sim p}\nabla \theta f(\theta,x)$

Variables: First term = positive phase, second term = negative phase
Conditions: Negative phase estimated via MCMC samples / 负相位通过MCMC样本估计

3.17 Langevin MCMC / 朗之万MCMC

Formula: $xt+1=xt+\epsilon \nabla f(\theta_t)x+\epsilon z_{2\epsilon t}$, where $z_t\sim N(0,1)$

Variables: ϵ = step size, $\nabla f(\theta_t)$ = score function

Conditions: Converges to $p\theta$ as $\epsilon\rightarrow 0$, $T\rightarrow\infty$ / 当步长趋于0，迭代次数趋于无穷时收敛

4. Typical Questions & Solutions / 典型例题与解答

Q1: Gaussian Filter Properties / 高斯滤波性质

Problem (问题): If you convolve an image with a Gaussian of σ_1 and then with a Gaussian of σ_2 , what is the effective σ ?
Solution (解答): $\sigma_{\text{eff}}=\sigma_1+2\sigma_2$. This follows from the property that convolution of two Gaussians is another Gaussian, and their variances add. / 两个高斯卷积仍为高斯，方差相加。

Q2: Nyquist Rate / 奈奎斯特率

Problem (问题): An image has maximum frequency 50 cycles/pixel. What is the minimum sampling rate to avoid aliasing?
Solution (解答): Nyquist rate is $2\times$ max frequency = 100 samples/pixel interval. Sampling rate must be ≥ 100 . / 采样率必须至少为最高频率的两倍。

Q3: Harris Corner Classification / Harris角点分类

Problem (问题): For a window where the second moment matrix H has eigenvalues $\lambda_1=5000$ and $\lambda_2=50$, what type of region is this?
Solution (解答): $\lambda_1\gg\lambda_2\rightarrow$ This is an edge region. The intensity changes significantly in one direction but not the other. / 这是一个边缘区域，强度在一个方向变化大但在另一方向变化小。

Q4: Homography DOF / 单应变换自由度

Problem (问题): How many point correspondences are needed to solve for a homography?
Solution (解答): A homography has 8 DOF (3x3 matrix up to scale). Each point correspondence gives 2 equations. Therefore, minimum 4 point correspondences are needed. / 单应矩阵有8个自由度，每对点提供2个方程，至少需4对点。

Q5: RANSAC Rounds / RANSAC轮数计算

Problem (问题): For computing a homography ($s=4$) with 30% outliers and $p=0.99$, how many RANSAC rounds?
Solution (解答): $N=\log(1-(0.99)^p)/\log(1-(1-0.3)^4)=\log(0.01)/\log(1-0.74)\approx 17$ rounds. / 约需17轮迭代。

Q6: Stereo Depth Calculation / 立体深度计算

Problem (问题): Given a stereo system with baseline $B=0.2\text{m}$, focal length $f=500$ pixels, and disparity $D=50$ pixels, what is the depth Z ?
Solution (解答): $Z=f\cdot B/d=500\times 0.2/50=2$ meters. / 深度为2米。

Q7: Conv Layer Output Size / 卷积层输出尺寸

Problem (问题): Input $32\times 32\times 3$, 10 filters of size $5\times 5\times 3$, stride 1, padding 0. What is the output size?
Solution (解答): $\text{Output}=(32-5+0)/1+1=28$. So output size is $28\times 28\times 10$. / 输出尺寸为 $28\times 28\times 10$ 。

Q8: VAE ELBO Decomposition / VAE的ELBO分解

Problem (问题): Write the ELBO in terms of reconstruction loss and KL divergence.
Solution (解答):

$\text{ELBO}=\mathbb{E}_q(z|x)[\log p(z|x;\theta)]-DKL(q(z|x)\parallel p(z))$

- First term: reconstruction loss (autoencoding) / 重建损失
- Second term: KL divergence of approximating posterior from prior / 近似后验与先验的KL散度

Q9: Flow Model Likelihood / 标准流化似然计算

Problem (问题): In a normalizing flow with 3 invertible layers, write the log-likelihood.
Solution (解答): $\log p(x;\theta)=\log Z(f^{-1}(x))=2\pi m-13\log|\det(\partial f_m/\partial z_m)|$

where $z_0=f^{-1}(x)$ and $z_m=f_m(z_{m-1})$. / 对数似然等于先验对数概率加上各层雅可比行列式对数之和。

Q10: GAN Optimal Discriminator / GAN最优判别器

Problem (问题): Derive the optimal discriminator for fixed generator.
Solution (解答):

$D^*(x)=p_{data}(x)/p_{data}(x)+p_G(x)$

- When $p_G=p_{data}$: $D^*=1/2$ (cannot distinguish) / 无法区分真假
- When $p_G\neq p_{data}$: D^* assigns higher probability to whichever distribution has higher density at x . / 判别器给密度更高的分布更高的概率。

Q11: Reparameterization Trick / 重参数化技巧

Problem (问题): Why can't we directly backpropagate through sampling $z\sim q(z|\text{varphi})$?
Solution (解答): Sampling is a stochastic operation – gradients cannot flow through random nodes. Reparameterization: $z=z_0+\epsilon$, where $\epsilon\sim N(0,I)$. Now z is a deterministic function of varphi and ϵ ; varphi's gradients can flow through μ and ϵ . / 采样是随机操作，梯度无法通过。重参数化将随机性转移到 ϵ ，使梯度可通过确定函数 μ 和 ϵ 传递。

Q12: Change of Variables (1D) / 变量替换（一维）

Problem (问题): If $Z\sim U[0,2]$ and $X=4Z$, find $p(X)$.

Solution (解答): $h(X)=X/4$, $h'(X)=1/4$

$p(X)=p(Z(X/4))\cdot 1/4=(1/2)\times (1/4)=1/8$

for $x\in[0,8]$. This is uniform on $[0,8]$, as expected. / 结果为 $[0,8]$ 上的均匀分布。

Q13: NICE vs Real-NVP / NICE与Real-NVP区别

Problem (问题): What is the key difference between NICE and Real-NVP coupling layers?
Solution (解答): - NICE: $xd+1:nz=zd+1:n+m(z;1,d)$; $d(z;1)$ (volume-preserving). / 非线性耦合，体积保持。 - Real-NVP: $xd+1:nz=\alpha\cdot\text{NExp}(\alpha d(z;1,d))+\mu(z;1,d)$; $d(z;1)=\exp(Z\alpha)$ (non-volume-preserving). / 仿射耦合，非线性保持，表达能力更强。

Q14: MAF vs IAF / 掩码自回归流与逆自回归流

Problem (问题): When would you choose MAF over IAF?

Solution (解答): Choose MAF when you need efficient density evaluation (training by MLE), since its inverse ($x\rightarrow z$) is parallel. Choose IAF when you need efficient sampling (generation), since its forward ($z\rightarrow x$) is parallel. / MAF适合训练（需密度估计），IAF适合生成（需快速采样）。

(High-Speed Camera Projection Matrix / 相机投影矩阵

Problem (问题): Write the full projection from world coordinates to pixel coordinates.

Solution (解答):

$x_{\text{pixel}}=K[R|t]x_{\text{world}}$

where $K=[f_x,0,c_x,0,0,f_y,0,c_y,0,0,1]$ (intrinsics), $R=3\times 3$ rotation, $t=3\times 1$ translation (extrinsics). / 内参矩阵K，旋转R和平移t组成完整投影。

End of CS5187 Lecture Material / CS5187复习资料完

Q16: WGAN vs. Original GAN / WGAN与原始GAN对比

Problem (问题): Why is Wasserstein distance preferable to Jensen-Shannon divergence as a GAN training objective?

Solution (解答): - Jensen-Shannon divergence: When p_{data} and p_G have non-overlapping supports (common in high dimensions), DJS $\approx\log 2$ is constant – gradient provides no useful learning signal. / 当分布不重叠时JS散度为常数，梯度为零，无法提供学习信号。 - Wasserstein distance: Provides a meaningful, continuous gradient even when distributions don't overlap. The loss correlates with sample quality, enabling more stable training. / 即使分布不重叠也提供有意义的连续梯度，损失与样本质量相关。 - Practical benefit: WGAN loss decreases as sample quality improves; can be used as a training monitor. / WGAN损失随样本质量提升而下降，可作为训练监控指标。

Q17: CycleGAN Cycle Consistency / CycleGAN循环一致性

Problem (问题): Why is cycle consistency necessary in CycleGAN? What happens without it?
Solution (解答): - Without cycle consistency: The adversarial losses alone only ensure that $G(X)$ looks like domain Y and $F(Y)$ looks like domain X , but there is no guarantee that $F(G(X))=X$. The generators could map all inputs to a single mode in the target domain (mode collapse). / 仅靠对抗损失无法保证往返一致，生成器可能将所有输入映射到目标域的单一模式。 - With cycle consistency: $F(G(X))=X$ and $G(F(Y))=Y$ enforce that the translation preserves information content – the round trip must recover the original. / 循环一致性迫使翻译保留信息内容，往返必须恢复原始输入。 - Loss: $z_{cyc}=E[X\parallel F(G(X))-X\parallel]+E[Y\parallel G(F(Y))-Y\parallel]$ / 循环一致性损失用L1距离衡量往返差异。

Q18: EBM Gradient Interpretation / EBM梯度解释

Problem (问题): Explain the positive phase and negative phase in EBM maximum likelihood training.

Solution (解答):

$\nabla \log p(\theta_{train})=\nabla \theta f(\theta_{train})$ –positive phase- $E_x-p\theta[\nabla f(\theta,x)]$ –negative phase

- Positive phase (正相位): Pushes up θ at the training data location, making training data more probable. / 在训练数据位置推高 θ ，提高训练数据概率。
- Negative phase (负相位): Pushes down θ at locations where the model currently assigns high probability (model samples), reducing probability of non-data regions. / 在模型当前高概率位置拉低 θ ，降低非数据区域的概率。
- Contrastive divergence: Approximates the negative phase with MCMC samples: $\nabla \theta f(\theta_{train})-\nabla \theta f(\theta_{sample})$ / 用MCMC采样近似负相位。

Q19: Score Function in EBMs / EBM中的分数函数

Problem (问题): Why is the score function $\nabla \log p(\theta,x)$ tractable for EBMs?